SHARP GP2Y1010 Dust Sensor

1. Sensor Overview

The SHARP GP2Y1010 is an optical air quality sensor designed to detect very fine particles like dust, pollen, and cigarette smoke. The sensor detects these using an internal infrared (IR) LED and a photodiode, which are arranged diagonally across a hollow airflow chamber. When air enters the sensor's chamber, the internal IR LED illuminates it. Clean air allows the light to pass through without interference. If dust particles are present, they scatter the light toward the photodiode, which then measures the intensity of the scattering and produces an analog voltage proportional to the amount of dust.

2. Calibrate the sensor

Place your active sensor inside a sealed, clean container. Let it sit for 3 to 5 minutes so any trapped dust physically settles to the floor of the box. Watch the live voltage output on the serial monitor screen from the relevant Arduino code. It will slowly drop and eventually bottom out at a completely stable number. That lowest stable number is your calibrated V(threshold). The value we got as V(threshold) was 0.1V.

3. Taking the Dust Readings

Even though the sensor is powered by 5V, the SHARP datasheet shows its analog output rarely goes above 3.5V even in extremely heavy dust. Because 3.5V is very close to the ESP32's 3.3V limit, we just wire it directly without a voltage divider to ESP32.

The IR LED inside the sensor cannot be left on continuously, as it draws too much power, heats up, and degrades. According to the manufacturer's datasheet, it must be pulsed on and off in strict 10-millisecond cycles. In the Arduino code, it manually controls this exact sequence using “delayMicroseconds()”.

The equations used;

Equation 1: Converting the Digital Number to Voltage

$$V_{pin} = Raw \times \left(\frac{3.3}{4095}\right)$$

Equation 2: The Linear Dust Conversion

The SHARP sensor does not output 0.0V in clean air. It has a natural resting voltage (often called the no-dust voltage). We took it as 0.1V.

$$Dust = \left(\frac{V_{pin} - V_{threshold}}{0.5}\right) \times 1000$$

- V(pin): The current voltage reading.
- V(threshold): The custom zero-dust baseline.
- 0.5: The datasheet voltage scaling factor.
- 1000: A multiplier used to scale the output into a more readable high-precision format (often mapping into (ug/m^3).